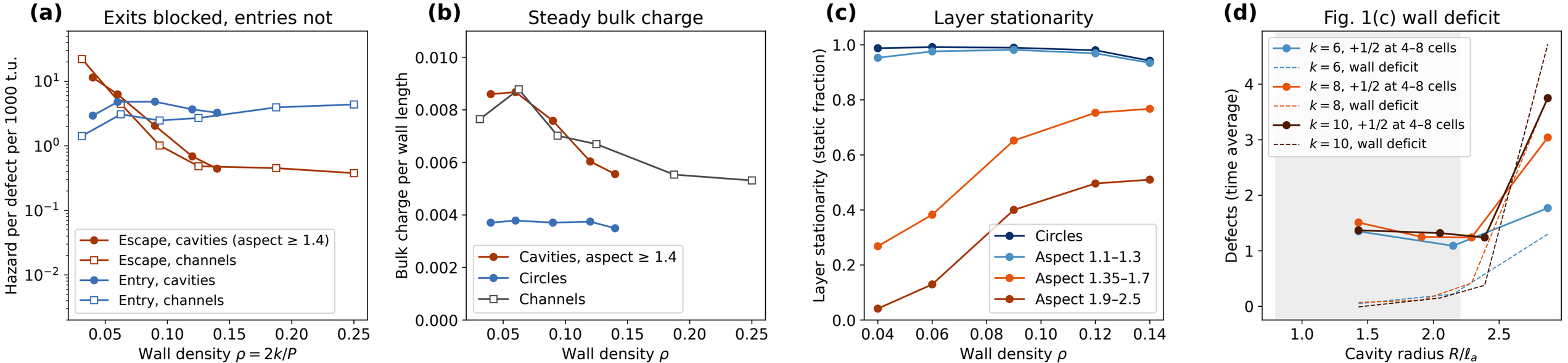
(a)
Exits blocked, entries not
Hazard per defect per 1000 t.u.
10^{1}
10^{0}
10^{-1}
10^{-2}
Escape, cavities (aspect ≥ 1.4)
Escape, channels
Entry, cavities
Entry, channels
0.05
0.10
0.15
0.20
0.25
Wall density $\rho = 2k/P$
(b)
Steady bulk charge
Bulk charge per wall length
0.010
0.008
0.006
0.004
0.002
0.000
Cavities, aspect ≥ 1.4
Circles
Channels
0.05
0.10
0.15
0.20
0.25
Wall density ρ
(c)
Layer stationarity
Layer stationarity (static fraction)
1.0
0.8
0.6
0.4
0.2
0.0
Circles
Aspect 1.1–1.3
Aspect 1.35–1.7
Aspect 1.9–2.5
0.04
0.06
0.08
0.10
0.12
0.14
Wall density ρ
(d)
Fig. 1(c) wall deficit
Defects (time average)
4
3
2
1
0
$k = 6$, +1/2 at 4–8 cells
$k = 6$, wall deficit
$k = 8$, +1/2 at 4–8 cells
$k = 8$, wall deficit
$k = 10$, +1/2 at 4–8 cells
$k = 10$, wall deficit
1.0
1.5
2.0
2.5
Cavity radius R/ℓ_a